Spectral analysis and block convolution

Answers and worked solutions · Day 11 · Tue 3 Nov · pack D6

5. Practice

From your course files

T5 Q5 (tutorial).

> *Hint: Playbook A.* $N \geq 6000/50$, *then round up to a power of 2.*
>
> (a) $F_s \geq 2B =$ **6 kHz**.
>
> (b) Resolution $= F_s/N \leq 50$ Hz, so $N \geq \dfrac{6000}{50} = 120$. $m = \log_2 N = 6.9$; rounding up to $m = 7$ gives $\mathbf{N = 2^7 = 128}$.
>
> (c) $T_0 = N/F_s = 128/6000 =$ **0.0213 s**. (The actual resolution is $6000/128 = 46.9$ Hz.)

T5 Q4 (tutorial).

> The inverse system is $H_I(z) = 1 - 0.5z^{-1}$, with $h_I(n) = \{\underset{\uparrow}{1}, -0.5\}$, so $x(n) = h_I(n) * y(n)$.
>
> 1. Zero-pad h_I to length 64: $\bar{h}_I(n) = \{h_I(n), 0, \ldots, 0\}$ (62 zeros). $y(n)$ is already length 64.
> 2. Compute $\bar{H}_I(k)$ and $Y(k)$ (64-point DFTs), then $\bar{h}_I(n) \circledast_{64} y(n) = \mathrm{IDFT}_{64}\{\bar{H}_I(k)Y(k)\}$.
> 3. For $n = 1, \ldots, 63$: $\bar{h}_I \circledast_{64} y = h_I(0)y(n) + h_I(1)y(n-1)$, which is the linear convolution, so it equals $x(n)$.
> 4. For $n = 0$ the circular result is $h_I(0)y(0) + h_I(1)y(63) = y(0) - 0.5y(63)$, but the linear one is $h_I(0)y(0)$ (with $y(n) = 0$ for $n < 0$). Correct it: $\mathbf{x(0) = \left[\bar{h}_I \circledast_{64} y\right]_{n=0} + 0.5\, y(63)}$.
>
> This is overlap-save with $M = 2$: the circular convolution of a length-N block with a length-M filter corrupts exactly the first $M - 1 = 1$ output.

T5 Q6 (tutorial).

> (a) $X(\omega) = \sum_{n=-2}^{3} x(n)e^{-j\omega n} = \mathbf{3 + 4\cos\omega + 2\cos(2\omega)}$ (the pairs $2e^{\pm j\omega}$ and $1 \cdot e^{\pm j2\omega}$ combine by Euler; $x(3) = 0$).
>
> (b) $V(k) = \sum_{n=0}^{5} v(n)e^{-j2\pi kn/6} = 3 + 4\cos\dfrac{\pi k}{3} + 2\cos\dfrac{2\pi k}{3} = \mathbf{\{9, 4, 0, 1, 0, 4\}}$.

(c) $\mathbf{V(k) = X(\omega)|_{\omega=2\pi k/6}}$: $V(k)$ samples $X(\omega)$ at $N = 6$ equally spaced frequencies. In the time domain, sampling $X(\omega)$ corresponds to the periodic extension $x_p(n) = \sum_l x(n - lN)$ with $N = 6$, i.e. $\{\ldots, 1, 2, \underset{\uparrow}{3}, 2, 1, 0, 1, 2, 3, \ldots\}$, and one period of it starting at $n = 0$ is $v(n) = \{\underset{\uparrow}{3}, 2, 1, 0, 1, 2\}$.

New questions

N1.

a. $F_s/N = 1000/200 = 5$ Hz. $T_0 = N/F_s = 200/1000 = 0.2$ s.

b. $k_0 = 120/5 = 24$, and its mirror image is $N - 24 = 176$. **Bins 24 and 176.** The tone is bin-aligned, so a rectangular window gives no leakage.

c. $k = 37$: $F = 37 \times 5 =$ **185 Hz**. $k = 150 > N/2 = 100$: it represents $-(200 - 150) \times 5 = -250$ **Hz**. For a real signal, it is the mirror image of bin 50 (250 Hz), not "750 Hz". 750 Hz is above $F_s/2$, so it cannot be present after anti-aliasing.

N2.

a. Bin spacing $F_s/N = 1000/512 = 1.95$ Hz. Resolution $1/T_0$ with $T_0 = L/F_s = 100/1000 = 0.1$ s, so the resolution is **10 Hz**.

b. The separation is 5 Hz, which is less than 10 Hz, so **no**. Zero-padding to 4096 only makes the bin spacing finer (0.24 Hz). It interpolates the same smeared DTFT, so the two tones stay as one lobe. **No.**

c. Need $1/T_0 \leq 5$ Hz, so $T_0 \geq 0.2$ s and $L \geq 0.2 \times 1000 =$ **200 samples**. The next power of 2 is $N = L = 256$ (resolution 3.9 Hz).

N3.

a. $M = 2$, so $N = L + M - 1 = 4$, and $\bar{h}(n) = \{1, -1, 0, 0\}$.

b. Blocks $x_0 = \{1, 0, 2\}$ and $x_1 = \{1, 3, 1\}$, each padded with one zero. A length-3 block convolved with length-2 h is 4 long, so the 4-point circular convolution equals the linear convolution:

- $y_0 = \{1, 0, 2\} * \{1, -1\} = \{1, -1, 2, -2\}$
- $y_1 = \{1, 3, 1\} * \{1, -1\} = \{1, 2, -2, -1\}$

c. y_0 is at $n = 0..3$ and y_1 at $n = 3..6$. They overlap by $M - 1 = 1$ sample at $n = 3$: $-2 + 1 = -1$.

$$y(n) = \{\underset{\uparrow}{1}, -1, 2, -1, 2, -2, -1\}$$

Check at $n = 3$: $y(3) = x(3) - x(2) = 1 - 2 = -1$. This matches. Length $6 + 2 - 1 = 7$, also correct.

N4.

a. $M = 3$, so $L = N - M + 1 = 5 - 3 + 1 = \mathbf{3}$.

b. Each block is $M - 1 = 2$ old samples followed by 3 new ones. The first block starts with 2 zeros. The full output has length $8 + 3 - 1 = 10$, so the number of blocks is $\lceil 10/3 \rceil = 4$:

- $x_0 = \{0, 0, 1, 3, 2\}$
- $x_1 = \{3, 2, 1, 0, 2\}$
- $x_2 = \{0, 2, 4, 1, 0\}$ (the last new sample is a padding zero)
- $x_3 = \{1, 0, 0, 0, 0\}$

4 blocks.

c. Linear convolution with h gives length 7. Fold the last 2 samples onto the first 2:

- $x_0 * h = \{0, 0, 2, 5, 2, \mathbf{1}, \mathbf{2}\}$, so $x_0 \circledast_5 h = \{0 + 1, 0 + 2, 2, 5, 2\} = \{1, 2, 2, 5, 2\}$
- $x_1 * h = \{6, 1, 3, 1, 5, \mathbf{-2}, \mathbf{2}\}$, so $x_1 \circledast_5 h = \{6 - 2, 1 + 2, 3, 1, 5\} = \{4, 3, 3, 1, 5\}$

Discard the first $M - 1 = 2$ outputs of each block ($\{1, 2\}$ and $\{4, 3\}$). They contain wrapped-around tail terms (time aliasing). Keep $\{2, 5, 2\}$ and $\{3, 1, 5\}$.

d. The other blocks: $x_2 \circledast_5 h = \{1, 4, 6, 0, 3\}$, so keep $\{6, 0, 3\}$.
$x_3 \circledast_5 h = \{2, -1, 1, 0, 0\}$, so keep $\{1, 0, 0\}$ and use only the first value.

$$y(n) = \{\underset{\uparrow}{2}, 5, 2, 3, 1, 5, 6, 0, 3, 1\}$$

Check: $y(6) = 2x(6) - x(5) + x(4) = 8 - 2 + 0 = 6$. This matches.

Overlap-add needs $\lceil 8/3 \rceil = \mathbf{3\ blocks}$ (the last block's tail supplies $y(9)$). Overlap-save needs one extra block here because each block delivers only L outputs.

N5.

a. **Hanning.** The weak tone is 40 dB down. The rectangular window's leakage never gets below about -26 dB within 6 bins, so it would bury the weak tone. The strong tone is not bin-aligned in general, so you cannot rely on zero leakage. Hanning's side lobes drop below -50 dB by 5 bins. The price is a wider main lobe, which is why (b) asks for 5 bins of separation.

b. Separation 40 Hz $\geq 5F_s/L$, so $L \geq 5 \times 8000/40 = 1000$. Round up: $L = 1024$, $T_0 = 1024/8000 =$ **0.128 s** (bin spacing 7.81 Hz, so the tones are 5.12 bins apart).

c. Rectangular, $L = 1024$: the leakage from 1010 Hz at the weak tone's bins is about -25 dB, which is above the weak tone (-40 dB). **No separate peak is visible.**

Numerical check (Python, $L = 1024$): with Hanning, the weak tone's bin is at -38.4 dB and its neighbours are at -49.5 and -43.1 dB, so there is a clear local peak. With rectangular, the same bin is at -25.9 dB with neighbours at -22.8 and -25.4 dB, so there is no peak. With Hanning and $L = 512$ (2.56 bins apart), there is no peak either.

N6.

a. $2B = 10$ kHz ≤ 12 kHz. **Yes.**

b. $\Delta F = 30$ Hz, so $N \geq 12000/30 = 400$. Round up to $N = 512$. $T_0 = 512/12000 =$ **42.7 ms**. Resolution $12000/512 =$ **23.4 Hz** ≤ 30 Hz.

c. Bin spacing 23.4375 Hz. $k_1 = 1500/23.4375 =$ **64** (bin-aligned, so no leakage on its own). $k_2 = 1530/23.4375 =$ **65.28** (not aligned, so it leaks into all bins, with its peak near $k = 65$). The mirror images are at $512 - 64 = 448$ and $512 - 65.28 = 446.72$. Caveat: $1/T_0 = 23.4$ Hz is the course's rule, and by that rule the tones count as resolved. On a real plot, though, the two tones sit only 1.28 bins apart, so you may not see a clear dip between two peaks. In practice you want a margin above $1/T_0$.

d. Bin spacing $12000/2048 =$ **5.86 Hz**. Resolution $1/T_0 = F_s/L = 12000/128 =$ **93.75 Hz**, which is more than 30 Hz. **No.** The tones merge into one lobe. (Python: the zero-padded spectrum has one main peak, at about 1516 Hz.) Padding only interpolates.

e. $L \geq 2F_s/\Delta F = 2 \times 12000/30 = 800$. Round up to $L = 1024$.
$T_0 = 1024/12000 =$ **85.3 ms**. (Python check: with Hanning and $L = 1024$, there are two peaks near 1500 and 1530 Hz.)

6. Quick recall check

Question

1. $2000/256 = 7.81$ Hz; $T_0 = 256/2000 = 0.128$ s.
2. No. It samples the same DTFT more finely (smaller bin spacing), which is interpolation only. The resolution is still $1/T_0$.
3. $w(n) = \frac{1}{2}(1 - \cos(2\pi n/4))$, so $w = \{0, 0.5, 1, 0.5, 0\}$.
4. $L = N - M + 1 = 13$ new samples. Discard the first $M - 1 = 3$ outputs of each block.

5. Hanning has much lower side lobes (-31 dB vs -13 dB, so less leakage) but a main lobe about twice as wide (worse resolution).